Creación de un Clúster en MongoDB Atlas e importación de una colección local.

Materia:
Base de Datos NoSQL

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Creación del Clúster

Primero deberemos instalar la CLI de Atlas en nuestro sistema (*CachyOS*):



Una vez instalado, lo iniciamos con:

```
atlas setup
```

Nos pedirá ingresar con nuestra cuenta de MongoDB Atlas; Una vez realizado ese proceso, nos preguntará como deseamos crear nuestro clúster:



- La opción default crea el clúster de manera automática, con los valores generados por la CLI.
- La opción custom, pedirá el nombre, el proveedor de servicios en la nube, el usuario *admin* de ese cluster, una contraseña y la región a de la nube.

Seleccionaremos la segunda opción y crearemos un cluster llamado: *cesarCluster*, el *Cloud Provider* será AWS, la región será US_EAST_1

```
→ atlas setup

[Default Settings]
Cluster Name:                          Cluster93196
Cloud Provider and Region:             AWS - US_EAST_1
Database User Username:                Cluster93196
Load sample data:                      Yes
Allow connections from (IP Address):   187.192.100.108

? How do you want to set up your Atlas cluster? custom
Press [Enter] to use the default values.

Enter [?] on any option to get help.

[Set up your Atlas cluster]
? Cluster Name [This can't be changed later] cesarCluster
? Cloud Provider AWS
? Cloud Provider Region US_EAST_1
? Do you want to load a sample dataset? No

[Set up your database authentication access details. Store them in a secure location.]
X Sorry, your reply was invalid: a user with this username cesar already exists
? Database User Username cesar
? Database User Password [You must include >10 characters. Press Enter to use an auto-generated password] cesar:v1212

[Set up your database network access details]
? IP Access List Entry [Press Enter to use your public IP address '187.192.100.108'] 187.192.100.108
Detected clusterWideScaling mode. If you require more flexibility in shard scaling, consider using --autoScalingMode independentShardScaling.
We are deploying cesarCluster...

Store your database authentication access details in a secure location:
Database User Username: cesar
Database User Password: cesar:v1212

Creating your cluster... [It's safe to 'Ctrl + C']
/
```

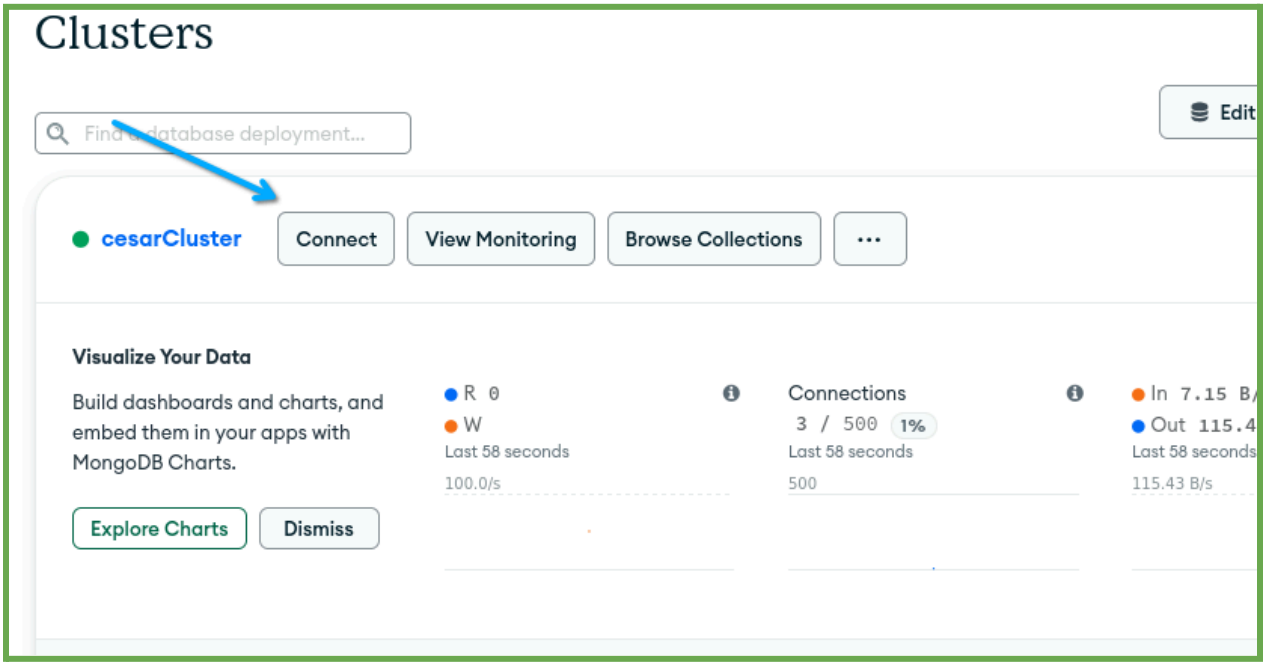
Una vez creado, no preguntará con qué queremos hacer la primera conexión, nosotros seleccionaremos *mongosh*:

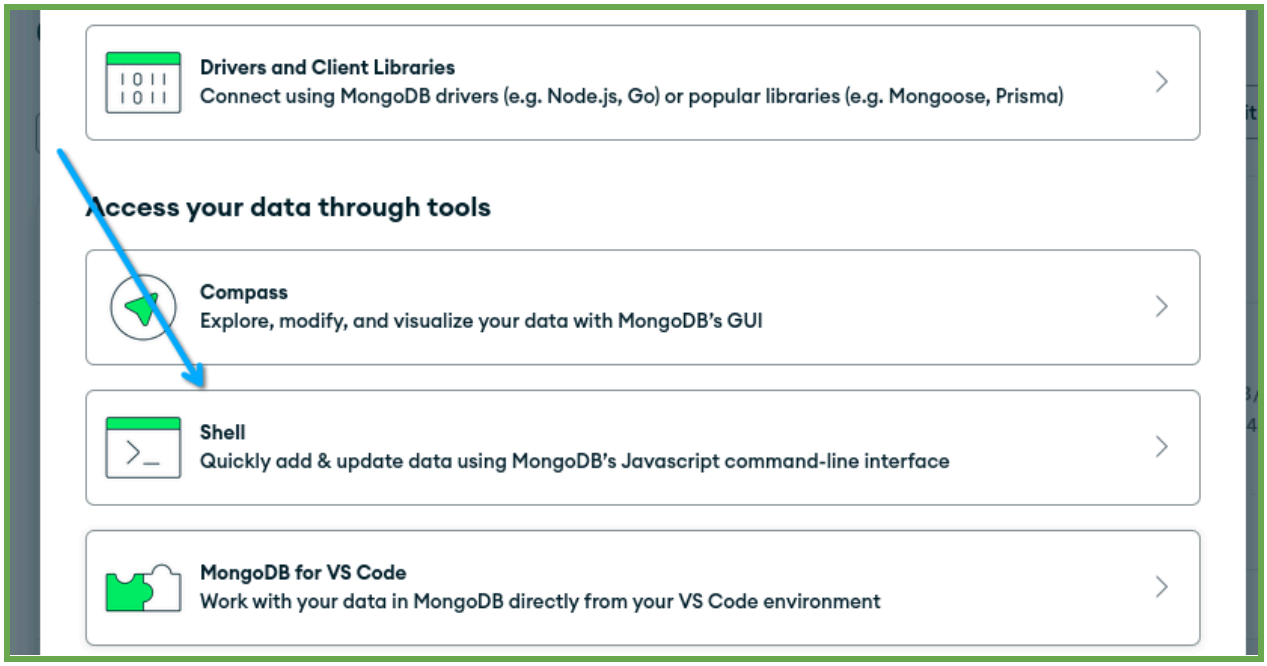
```
? How would you like to connect to your cluster? mongosh
Current Mongosh Log ID: 6ab73c5ad8d948fab984be60
Connecting to:          mongodb+srv://<credentials>@cesarcluster.vbh3mwk.mongodb.net/?appName=mongosh+2.12.0
Using MongoDB:          8.0.32
Using Mongosh:          2.12.0

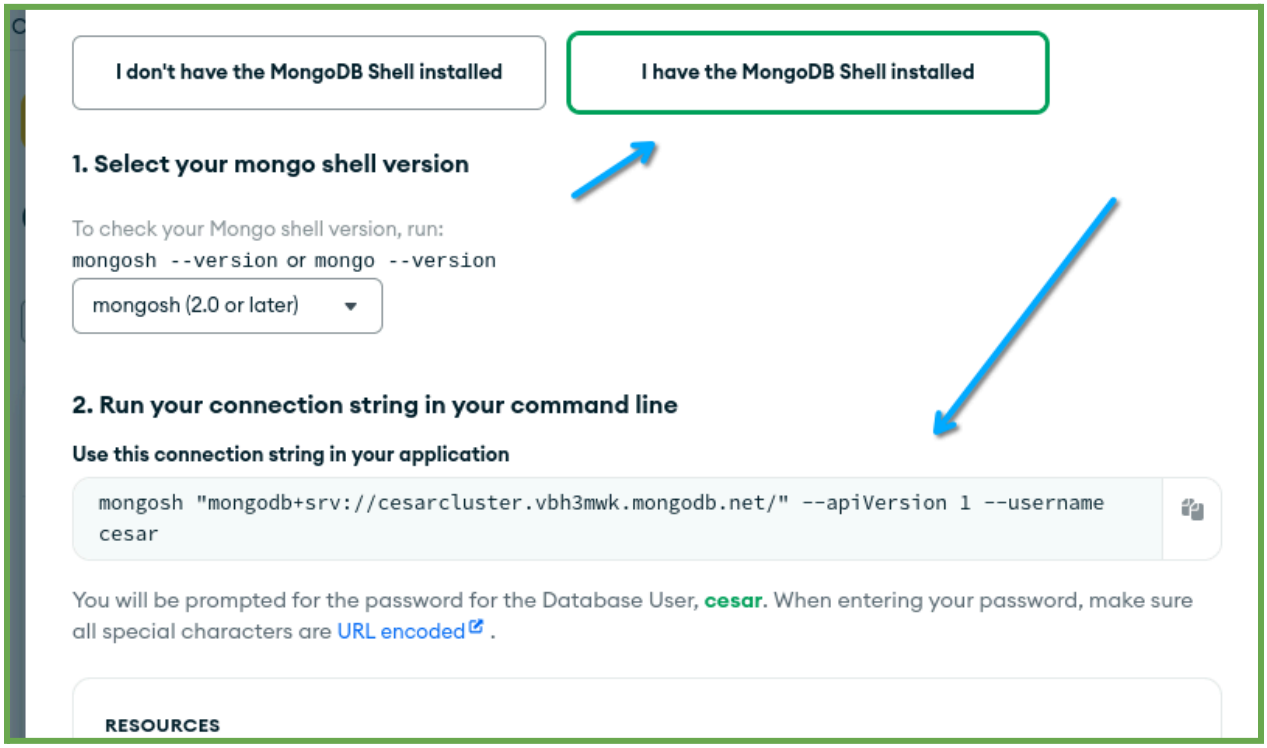
For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

Atlas atlas-nrbgml-shard-0 [primary] test>
```

De igual manera el *string de conexión ó URI* se puede encontrar en el dashboard de MongoDB Atlas:




Drivers and Client Libraries
Connect using MongoDB drivers (e.g. Node.js, Go) or popular libraries (e.g. Mongoose, Prisma)
Access your data through tools
Compass
Explore, modify, and visualize your data with MongoDB's GUI
Shell
Quickly add & update data using MongoDB's Javascript command-line interface
MongoDB for VS Code
Work with your data in MongoDB directly from your VS Code environment


I don't have the MongoDB Shell installed
I have the MongoDB Shell installed
1. Select your mongo shell version
To check your Mongo shell version, run:
mongosh --version or mongo --version
mongosh (2.0 or later)
2. Run your connection string in your command line
Use this connection string in your application
mongosh "mongodb+srv://cesarcluster.vbh3mwk.mongodb.net/" --apiVersion 1 --username cesar
You will be prompted for the password for the Database User, cesar. When entering your password, make sure all special characters are URL encoded .
RESOURCES

Importación de un colección local.

En este caso se importará un colección de 540 registros, que se encuentra en un contenedor de Podman, para lo cual:

```
cesar ~/Documentos/ia-datos v26.8.2 21:40
→ podman exec -it spam-mongo bash
mongodb@9ab623bfb969:/$
```

Una vez en la sesión de Bash, ejecutaremos el comando mongodump, le especificamos que colección queremos respaldar y en donde se guarda el respaldo (/tmp/mongo_backup porque estamos en un contenedor):

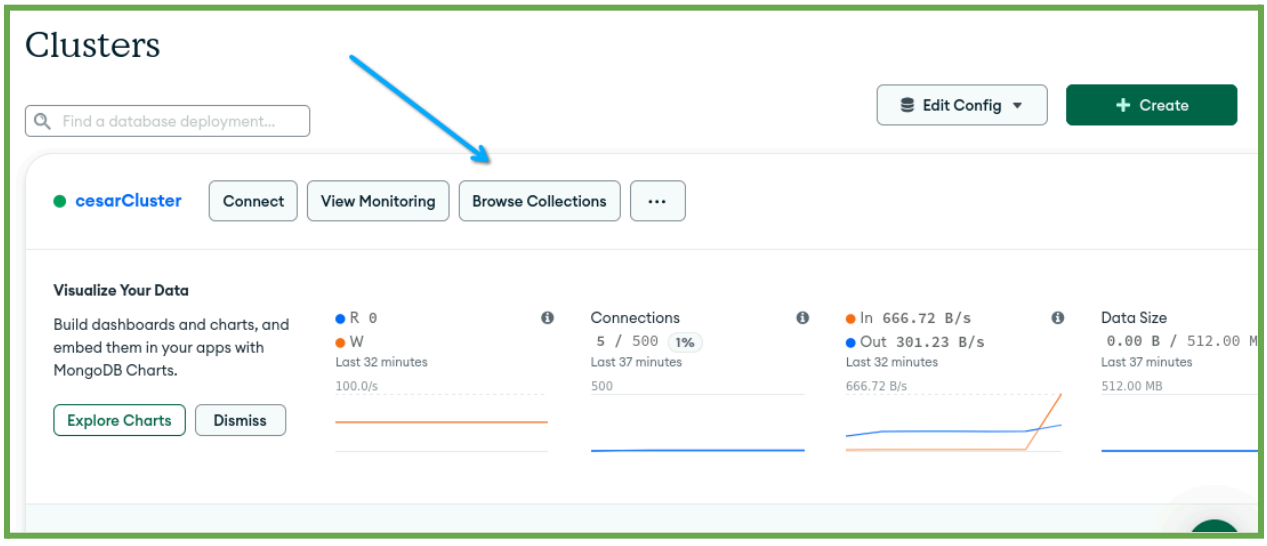
```
mongodb@9ab623bfb969:/$ mongodump --host localhost --port 27017 --db emails_db --collection emails --out /tmp/mongo_backup
2026-09-26T03:48:18.720+0000    writing `emails_db.emails` to `/tmp/mongo_backup/emails_db/emails.bson`
2026-09-26T03:48:18.723+0000    done dumping `emails_db.emails` (540 documents)
mongodb@9ab623bfb969:/$
```

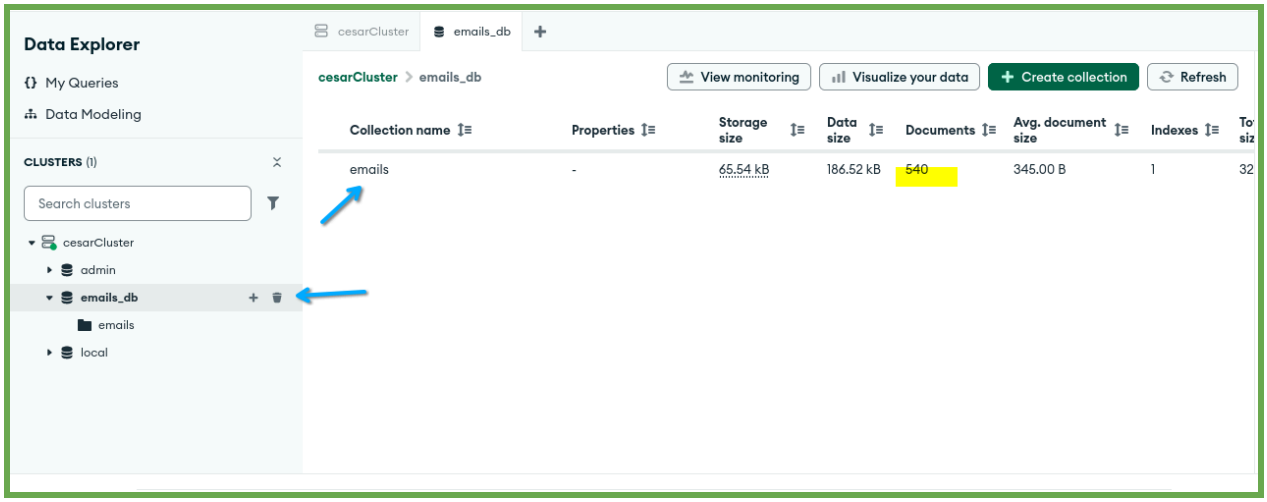
Ahora usaremos la utilidad mongorestore y la URI de conexión para montarlo en el cluster de Atlas, debemos especificar la BD y la colección, así como pasarle el archivo .bson:

```
mongodb@9ab623bfb969:/$ mongorestore --uri "mongodb+srv://cesarcluster.vbh3mwk.mongodb.net/emails_db" --username cesar --collection=emails /tmp/mongo_
backup/emails_db/emails.bson
Enter password for mongo user:

2026-09-26T04:02:52.967+0000    checking for collection data in `/tmp/mongo_backup/emails_db/emails.bson`
2026-09-26T04:02:52.967+0000    reading metadata for `emails_db.emails` from `/tmp/mongo_backup/emails_db/emails.metadata.json`
2026-09-26T04:02:53.138+0000    restoring `emails_db.emails` from `/tmp/mongo_backup/emails_db/emails.bson`
2026-09-26T04:02:55.479+0000    finished restoring `emails_db.emails` (540 documents, 0 failures)
2026-09-26T04:02:55.479+0000    no indexes to restore for collection `emails_db.emails`
2026-09-26T04:02:55.479+0000    540 document(s) restored successfully. 0 document(s) failed to restore.
mongodb@9ab623bfb969:/$
```

Ahora solo checamos en el dashboard de Atlas:




Data Explorer
My Queries
Data Modeling
CLUSTERS (1)
Search clusters
cesarCluster
admin
emails_db
emails
local
cesarCluster
emails_db
cesarCluster > emails_db
View monitoring
Visualize your data
Create collection
Refresh
Collection name
Properties
Storage size
Data size
Documents
Avg. document size
Indexes
emails
-
65.54 kB
186.52 kB
540
345.00 B
1
32